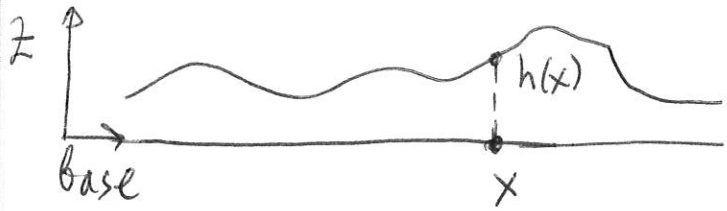
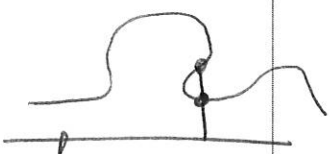
Statistical Mechanics of 2D surfaces (membranes)

1d surface in 2D : planar ~~line~~ curve



Can describe by height $h(x)$ above the base plane

Complications: surface with overhangs when $h(x)$ is multivalued.
Assume not relevant for now.



(1D) Describe curve by $z = h(x)$. Arbitrary point on a curve

$$\vec{R} = (x, h(x)) = x\hat{e}_x + h(x)\,\hat{e}_z$$

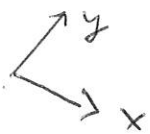
$$d\vec{R} = \frac{d\vec{R}}{dx}\,dx = \left(\hat{e}_x + \frac{dh}{dx}\,\hat{e}_z\right)dx$$

length of segment $ds = (d\vec{R}\cdot d\vec{R})^{1/2} = \sqrt{1+\left(\frac{dh}{dx}\right)^2}\,dx$

length of curve $L = \int ds = \int_0^{L_B} dx\,\sqrt{1+\left(\frac{dh}{dx}\right)^2}$

(2D)

$$z = h(x,y)$$



$$\vec{R} = (x, y, h(x,y)) = x\hat{e}_1 + y\hat{e}_2 + h\hat{e}_3$$

$$d\vec{R} = \frac{\partial\vec{R}}{\partial x}\,dx + \frac{\partial\vec{R}}{\partial y}\,dy$$

$$\frac{\partial\vec{R}}{\partial x} = \hat{e}_1 + \frac{\partial h}{\partial x}\,\hat{e}_3$$

$$\frac{\partial\vec{R}}{\partial y} = \hat{e}_2 + \frac{\partial h}{\partial y}\,\hat{e}_3$$

Area $d\vec{A} = \frac{\partial \vec{R}}{\partial x} dx \otimes \frac{\partial \vec{R}}{\partial y} dy = \frac{\partial \vec{R}}{\partial x} \times \frac{\partial \vec{R}}{\partial y} dx dy$

$$\frac{\partial \vec{R}}{\partial x} \times \frac{\partial \vec{R}}{\partial y} = \begin{vmatrix} \hat{e}_1 & \hat{e}_2 & \hat{e}_3 \\ 1 & 0 & \partial_x h \\ 0 & 1 & \partial_y h \end{vmatrix} = -\partial_x h\, \hat{e}_1 - \partial_y h\, \hat{e}_2 + \hat{e}_3$$

$d\vec{A}$ is oriented normal to the surface

$\Rightarrow$ normal unit vector $\vec{N} \sim d\vec{A}$

$$\vec{N} = \frac{d\vec{A}}{|dA|}$$

$$|d\vec{A}| = \left(d\vec{A} \cdot d\vec{A}\right)^{1/2} = \sqrt{(\partial_x h)^2 + (\partial_y h)^2 + 1}\; dx dy$$

Define $\vec{\nabla} h = \frac{\partial h}{\partial x} \hat{e}_1 + \frac{\partial h}{\partial y} \hat{e}_2$

$$\Rightarrow (\partial_x h)^2 + (\partial_y h)^2 = (\vec{\nabla} h)^2$$

$$dA = \sqrt{1 + (\vec{\nabla} h)^2}\; dx dy$$

$$\vec{N} = \frac{\left(-\partial_x h\, \hat{e}_1 - \partial_y h\, \hat{e}_2 + \hat{e}_3\right) dx dy}{\sqrt{1 + (\vec{\nabla} h)^2}\; dx dy} =$$

$$= \frac{-\partial_x h\, \hat{e}_1 - \partial_y h\, \hat{e}_2 + \hat{e}_3}{\sqrt{1 + (\vec{\nabla} h)^2}} \approx \hat{e}_3 - \vec{\nabla} h(x,y)$$

for flat surface $h = \text{const} \Rightarrow \vec{N} = \hat{e}_3$.

Elastic energy of bending deformation

$$E_0 = \frac{1}{2}\varkappa \int d^2\vec{r}\,(\vec{\nabla}\cdot\vec{N})^2 \approx \frac{1}{2}\varkappa \int dx\,dy\,(\nabla^2 h)^2$$

$$\downarrow$$

$$\left(\hat{e}_1 \frac{\partial}{\partial x} + \hat{e}_2 \frac{\partial}{\partial y}\right)$$

Surface tention contribution $E_1 = \int dA\,\sigma(\vec{N})\ (=\sigma A)$

i.e. energy proportional to area A,

σ = surface tension coefficient.

Note $dA = \sqrt{1+(\vec{\nabla}h)^2}\,dx\,dy \approx \left(1+\frac{1}{2}(\vec{\nabla}h)^2\right)dx\,dy$

$$\Rightarrow\ E = E_0 + E_1 = \text{const} + \int dx\,dy\left\{\frac{\sigma}{2}(\vec{\nabla}h)^2 + \frac{\varkappa}{2}(\vec{\nabla}^2 h)^2\right\}$$

Dynamics of height field comes from kinetic energy term $T = \frac{1}{2}\int dx\,dy\ \rho\left(\frac{\partial h}{\partial t}\right)^2$, where ρ = mass density.

Hence classical lagrangian

$$L = T - V = \int d^2\vec{r}\left[\frac{\rho}{2}\left(\frac{\partial h}{\partial t}\right)^2 - \left\{\frac{\sigma}{2}(\vec{\nabla}h)^2 + \frac{\varkappa}{2}(\vec{\nabla}^2 h)^2\right\}\right]$$

E-L eqns: $\frac{\partial}{\partial t}\frac{\delta L}{\partial \dot{h}} - \frac{\partial L}{\partial h} = 0$

$$\frac{\partial}{\partial t}\frac{\partial L}{\partial \dot{h}} = \rho\frac{\partial^2 h}{\partial t^2}$$

$$\frac{\partial L}{\partial h} = \sigma\vec{\nabla}^2 h - \varkappa\vec{\nabla}^2(\vec{\nabla}^2 h)$$

let $h(\vec{r},t) \sim e^{i\vec{k}\vec{r}-i\omega t}$

$$\Rightarrow\ -\rho\omega^2 = -\sigma k^2 - \varkappa k^4\ ;\ \omega = \sqrt{\frac{\sigma}{\rho}k^2 + \frac{\varkappa}{\rho}k^4}$$

Observe that in the absence of tention ($\sigma = 0$) the bending mode has dispersion $\omega = \sqrt{\frac{\varkappa}{\rho}}\, k^2$.

We get the same answer if we quantize displacement $h_r = \frac{1}{\sqrt{V}} \sum_k e^{i\bar{k}\bar{r}} h_k$ (see p. 9)

V = volume

Then from p. 11
$$\begin{cases} a_k = \sqrt{\frac{m\omega_k}{2\hbar}}\, h_k + \frac{i}{\sqrt{2\hbar m\omega_k}}\, p_k \\ a_k^+ = \sqrt{\frac{m\omega_k}{2\hbar}}\, h_{-k} + \frac{-i}{\sqrt{2\hbar m\omega_k}}\, p_{-k} \end{cases}$$

Hence $h_k = \sqrt{\frac{\hbar}{2m\omega_k}}\left(a_k + a_{-k}^+\right)$

Estimate typical displacement

$$\langle h_r \rangle = \frac{1}{\sqrt{V}} \sum_k \sqrt{\frac{\hbar}{2m\omega_k}} \langle (a_k + a_{-k}^+) \rangle e^{i\bar{k}\bar{r}} = 0$$

but $\langle h_r^2 \rangle \neq 0$

$$\langle h_r^2 \rangle = \frac{1}{V} \sum_{k_1 k_2} \frac{\hbar}{2m\sqrt{\omega_{k_1}\omega_{k_2}}}\, e^{i\bar{r}(\bar{k}_1 + \bar{k}_2)}$$

$$\cdot \langle \underbrace{(a_{k_1} + a_{-k_1}^+)(a_{k_2} + a_{-k_2}^+)}_{\langle a_{k_1} a_{-k_2}^+ \rangle + \langle a_{-k_1}^+ a_{k_2} \rangle =} \rangle$$

$$= \delta_{-k_2, k_1} \langle a_{k_1} a_{k_1}^+ \rangle + \delta_{-k_1, k_2} \langle a_{k_2}^+ a_{k_2} \rangle$$

$$\Rightarrow \langle h_r^2 \rangle = \frac{1}{V} \sum_k \frac{\hbar}{2\omega(k)} \left(1 + 2\langle a_k^+ a_k \rangle\right)$$

Since $\langle a_k^\dagger a_k \rangle = n_k = \dfrac{1}{e^{\beta\hbar\omega_k}-1}$

$$1 + 2n_k = 1 + \frac{2}{e^{\beta\hbar\omega_k}-1} = \frac{e^{\beta\hbar\omega_k}+1}{e^{\beta\hbar\omega_k}-1} = \coth\left(\frac{\beta\hbar\omega_k}{2}\right)$$

But $\dfrac{1}{V}\sum_k \cdots = \dfrac{1}{V} V \displaystyle\int \frac{d\vec{k}}{(2\pi)^d} \cdots$ (d = dimensionality)

$$\langle h_r^2 \rangle = \int \frac{d^2\vec{k}}{(2\pi)^2} \frac{\hbar}{2m\omega(k)} \coth\left(\frac{\hbar\omega(k)}{2k_BT}\right)$$

General result, valid for any $\omega(k)$.

Analyze limits: $\begin{cases} \coth\left(\frac{x}{2}\right) \xrightarrow[x \gg 1]{} 1 \\ \coth\left(\frac{x}{2}\right) = \dfrac{e^x+1}{e^x-1} \xrightarrow[x \ll 1]{} \dfrac{2}{x} \end{cases}$

$\Rightarrow$ For $\vec{k}$ such that $\hbar\omega(k) \gg k_BT$ we can replace $\coth(\,) \to 1$, then the integral

$$\sim \int \frac{d^2k}{\omega(k)} = \int^{\Lambda} \frac{dk \cdot k}{k^2} = \int \frac{dk}{k} = \ln(k).$$

For $\vec{k}$ such that $\dfrac{\hbar\omega(k)}{k_BT} \ll 1$ we instead get

$$\langle h_r^2 \rangle \cong \int^{k_T} \frac{d^2\vec{k}}{(2\pi)^2} \frac{\hbar}{2m\omega(k)} \frac{4k_BT}{\hbar\omega(k)} =$$

$$= \frac{2k_BT}{m} \frac{2\pi}{(2\pi)^2} \int^{k_T} \frac{dk \cdot k}{\left(\frac{\sigma}{\rho}k^2 + \frac{\varkappa}{\rho}k^4\right)}$$

Here k_T is determined from $\hbar\omega(k_T) = k_BT$

let's focus on $\underline{\sigma = 0}$ situation as the most dramatic one: $\hbar \frac{\varkappa}{\rho} k_T^2 = k_BT$

$$k_T \sim \sqrt{\frac{\rho k_BT}{\hbar\varkappa}}$$

$$\int_{k_0}^{k_T} \frac{dk}{k^3} = -\frac{1}{2}\int_{k_0}^{k_T} d\left(\frac{1}{k^2}\right) = -\frac{1}{2}\left(\frac{1}{k_T^2} - \frac{1}{k_0^2}\right)$$

$$\Rightarrow \langle h_r^2 \rangle \sim T\left(\frac{1}{k_0^2} - \frac{1}{k_T^2}\right)$$

Read: pp. 132-134 of Castro Neto et al Review

k_0 = lower cut-off which can be estimated as $k_0 \sim \frac{2\pi}{L}$ (L = linear size)

Since $k_0 \ll k_T$, $\frac{1}{k_0} \gg \frac{1}{k_T}$

$\Rightarrow \langle h_r^2 \rangle \sim T L^2 \Rightarrow$ membrane is highly distorted and corrugated!

Need tension $\sigma \neq 0$ to flatten it!